

Nimbus Edge-7 Sensor Hub — Technical Specification Sheet

Model: NE7-2026 | Revision: C

1. Overview

The Edge-7 Sensor Hub is a compact industrial IoT gateway designed to aggregate readings from up to 32 wired or wireless sensors and forward them to a cloud endpoint or local server over Ethernet, Wi-Fi, or LTE Cat-4. It is rated for continuous outdoor operation in temperatures from -20C to 60C.

2. Physical Specifications

Attribute	Value
Dimensions	142 mm x 96 mm x 34 mm
Weight	310 g
Enclosure Rating	IP67
Mounting	DIN rail or wall-mount bracket (included)
Operating Temperature	-20C to 60C
Storage Temperature	-40C to 85C

3. Connectivity

Interface	Details
Ethernet	1x RJ45, 10/100/1000 Mbps
Wi-Fi	802.11 b/g/n/ac, 2.4 GHz and 5 GHz
Cellular	LTE Cat-4, nano-SIM, requires separate data plan
Sensor Bus	RS-485 (Modbus RTU) and 2x digital I/O headers
Bluetooth	BLE 5.2, used for local configuration only

4. Power

The Edge-7 accepts 12-24V DC input via a locking barrel connector or terminal block, drawing a typical 4.2W at idle and up to 9.5W under full sensor and cellular load. An optional backup supercapacitor module (sold separately) provides up to 90 seconds of ride-through during power loss, enough to flush buffered readings to local flash storage before shutdown.

5. Data Handling

Sensor readings are buffered locally in 8 GB of onboard flash storage, holding approximately 45 days of data at a 1-minute polling interval across all 32 channels. If the network connection drops, the device automatically resumes syncing from the last acknowledged reading once connectivity is restored, avoiding data loss during outages of up to 45 days.

6. Certifications

CE, FCC Part 15 Class A, UL 61010-1, and RoHS 3 compliant. Cellular variants additionally carry PTCRB certification for use on major North American carrier networks.